The Master p-Adic Deformed Arithmetic Diamond Pencil, Bivariate Cross-Washout Containment, and Noisy Split-Mark Channel Capacity

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Abstract. We establish three exact analytical theorems across algebraic number theory, evolutionary mathematical biology, and statistical psychometrics: (1) **The Master p-Adic Deformed Arithmetic Diamond Pencil:** For any prime $p \geq 2$, exponent $r \geq 1$, and deformation parameter $\lambda \in \mathbb{C}$, the p-adic valuation-coupled GCD matrix $A_{n,p}^{(r)}(\lambda) = [\gcd(i,j)^r(1+\lambda\delta_{v_p(i),v_p(j)})]_{1\leq i,j\leq n}$ has exact determinant $\det A_{n,p}^{(r)}(\lambda) = \prod_{1\leq m\leq n,\, p\nmid m} J_r(m)^{V_{p,m}+1} p^{rV_{p,m}(V_{p,m}-1)/2} \Delta_{V_{p,m}}(p^r,\lambda)$, where $V_{p,m} = \lfloor \log_p(n/m) \rfloor$ and $\Delta_V(q,\lambda) = ((q+1)\lambda+q-1)\Delta_{V-1} - q\lambda^2\Delta_{V-2}$ ($\Delta_0 = 1+\lambda, \Delta_1 = q(1+\lambda)^2 - 1$) satisfies the universal Cassini identity $\Delta_V^2 - \Delta_{V+1}\Delta_{V-1} = \lambda^{2V} q^{V-1}$. Specializing (p,r,λ) unifies 16+ OEIS sequences across five algebraic regimes, including Pisot sequences $T(2,4q-1)$ (`A020727`, `A022026`, `A234279`), hollow Gaussian Lucas polynomials $(-1)^V U_V(-2,q)$ (`A009545`, `A088137`, `A088138`, `A045873`, `A088139`, `A098656`), and the Dual LCM Arithmetic Diamond Matrix $L_n^{(r)} = [i^r j^r / \gcd(i^r+j^r, |i^r-j^r|)]$ at $\lambda = -1/2$. (2) **Bivariate Poisson-Bernoulli Cross-Washout & $86.12\times$ Overlap Asymmetry:** During an adaptive therapy switch from primary drug $C_1(t) = C_{1,\text{off}} e^{-\gamma_1 t}$ to secondary collateral drug $C_2(t) = C_{2,\text{on}}(1-e^{-\gamma_2 t})$, the nonlinear crowding integral expands as an exact expectation over $(M_1, M_2) \sim \text{Poisson}(\Lambda_1) \times \text{Poisson}(-\Lambda_2)$, the normalized dual-exposure toxicity impulse obeys $\mathcal{I}_{2\to1}/\mathcal{I}_{1\to2} = (\gamma_1/\gamma_2)^2 = 86.1184$ between abiraterone (`3RUK`, $t_{1/2} = 15.0$ h) and enzalutamide (`2AM9`, $t_{1/2} = 5.8$ d), and 3-clone Volterra cycle integration yields the Double-Bind Extinction Wedge $\mu_1 < (1-1/\alpha_{21}) + \mu_2/\alpha_{21}$. (3) **Noisy Split-Mark Channel Capacity & Diachronic Clause-Bitrate Conservation:** Under any stochastic DINA channel with slip s and guess g, antipodal states $\eta = (1,0)$ and $\eta' = (0,1)$ have zero scalar-sum Kullback-Leibler divergence $D_{\text{KL}}(S_j) \equiv 0$ across all $(s,g) \in [0,1]^2$, whereas the ordered Split-Mark vector $V_j = (M_{j,L}, M_{j,R})$ achieves symmetric Jeffreys divergence $J(V_j) = 2(1-s-g)\log_2 \frac{(1-s)(1-g)}{sg}$ bits (15.29 bits at $s = g = 0.05$). Across paired 1800 to 2026 English prose, per-word syntactic coding efficiency η_{syn} surges by $+53.14\%$ ($p = 8.0 \times 10^{-6}$) while sentence-level propositional load remains conserved ($2.70 \to 2.92$, $p = 0.4778$).

1. The Master p-Adic Deformed Arithmetic Diamond Pencil $A_{n,p}^{(r)}(\lambda)$

Theorem 1.1 (Master p-Adic KMS Pencil & Universal Cassini Identity). Let $p \geq 2$ be prime, $r \geq 1$ an integer, and $\lambda \in \mathbb{C}$. Define the $n \times n$ matrix $A_{n,p}^{(r)}(\lambda) = \left[\gcd(i,j)^r(1+\lambda\delta_{v_p(i),v_p(j)})\right]_{1\leq i,j\leq n}$. By p-coprime Jordan totient Möbius congruence, $A_{n,p}^{(r)}(\lambda)$ is congruent over $\mathbb{Z}$ to a direct sum of $(V_{p,m}+1) \times (V_{p,m}+1)$ Kac-Murdock-Szegő (KMS) Toeplitz blocks indexed by $1 \leq m \leq n$ with $p \nmid m$ and $V_{p,m} = \lfloor \log_p(n/m) \rfloor$:

$$\det A_{n,p}^{(r)}(\lambda) = \prod_{\substack{m=1 \\ p\nmid m}}^{n} J_r(m)^{V_{p,m}+1}\, p^{rV_{p,m}(V_{p,m}-1)/2}\, \Delta_{V_{p,m}}(p^r,\lambda)$$

where $J_r(m) = m^r \prod_{\ell\mid m}(1-\ell^{-r})$ is Jordan's totient and $\Delta_V(q,\lambda)$ satisfies:

$$\Delta_0(q,\lambda) = 1+\lambda, \qquad \Delta_1(q,\lambda) = q(1+\lambda)^2 - 1, \qquad \Delta_V(q,\lambda) = \big((q+1)\lambda + q - 1\big)\Delta_{V-1}(q,\lambda) - q\lambda^2\Delta_{V-2}(q,$$

with exact generating function and universal Cassini identity for all $V \geq 1$:

$$\sum_{V=0}^{\infty} \Delta_V(q,\lambda) x^V = \frac{(1+\lambda) + (\lambda - q\lambda + q\lambda^2)x}{1 - ((q+1)\lambda + q - 1)x + q\lambda^2 x^2}, \qquad \big(\Delta_V(q,\lambda)\big)^2 - \Delta_{V+1}(q,\lambda)\Delta_{V-1}(q,\lambda) = \lambda^{2V} q^{V-1}.$$

Regime & Coupling λ	Polynomial $\Delta_V(q,\lambda)$ & Recurrence	Unified OEIS Sequences
Standard Diamond ($\lambda = 1, p = 2$) $A_n^{(r)} = [\gcd(i^r+j^r, \lvert i^r-j^r\rvert)]$	$\Delta_V(q,1) = 2q\Delta_{V-1} - q\Delta_{V-2}$ Pisot sequence $T(2,4q-1)$	`A020727` ($q=2$: 2, 7, 24, 82) `A022026` ($q=4$: 2, 15, 112, 836) `A234279` ($q=16$: 2, 63, 1984)

Regime & Coupling λ	Polynomial $\Delta_V(q,\lambda)$ & Recurrence	Unified OEIS Sequences
Hollow Valuation-Resonance ($\lambda=-1$) Zero diagonal ($\delta_{v_p(i),v_p(j)}=0$)	$\Delta_V(q,-1)=-2\Delta_{V-1}-q\Delta_{V-2}$ $(-1)^V U_V(-2,q)$, $\det A_n^{(r)}(-1)=-\delta_{n,2}$	`A009545` ($q=2$), `A088137` ($q=3$), `A088138` ($q=4$), `A045873` ($q=5$), `A088139` ($q=6$), `A098656` ($q=8$)
Dual LCM Diamond ($\lambda=-1/2, p=2$) $L_n^{(r)}=[i^r j^r/\gcd(i^r+j^r,\lvert i^r-j^r\rvert)]$	$\Delta_V(q,-1/2)=(-1)^V 2^{-(V+1)}N_V(q)$ $N_V=(3q-1)N_{V-1}-qN_{V-2}$ ($N_0=1, N_1=4q-1$)	Cassini: $N_V^2-N_{V+1}N_{V-1}=4q^{V+1}$ $N_V(2)=1,7,33,151,689,3143$
Sign-Resonance ($\lambda=-2$) $\gcd(i,j)^r(-1)^{\delta_{v_p(i),v_p(j)}}$	$\Delta_V(q,-2)=-(q+3)\Delta_{V-1}-4q\Delta_{V-2}$ $\Delta_0=-1, \Delta_1=q-1$	`A087170` ($q=4$: $-1,3,-5,-13,171$) `A370191` ($q=5$: $-1,4,-12,16,112$)

2. Bivariate Cross-Washout Trajectory & Double-Bind Extinction Wedge

Theorem 2.1 (Bivariate Poisson-Bernoulli Cross-Washout, Overlap Asymmetry & Extinction Wedge). Under an adaptive switch at $t=0$ from primary agent $C_1(t)=C_{1,\text{off}}e^{-\gamma_1 t}$ to secondary collateral agent $C_2(t)=C_{2,\text{on}}(1-e^{-\gamma_2 t})$, the sensitive burden $S_1(t)$ satisfies $S_1(t)=S_{1,\text{switch}}\Psi(t)/[1+(r_1 S_{1,\text{switch}}/K)\mathcal{J}(t)]$ where:

$$\mathcal{J}(t)=\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty}\frac{e^{-\Lambda_1}\Lambda_1^{m_1}}{m_1!}\frac{e^{\Lambda_2}(-\Lambda_2)^{m_2}}{m_2!}\frac{e^{(r_{\text{net}}-m_1\gamma_1-m_2\gamma_2)t}-1}{r_{\text{net}}-m_1\gamma_1-m_2\gamma_2},\qquad \frac{\mathcal{I}_{2\to1}}{\mathcal{I}_{1\to2}}=\left(\frac{\gamma_1}{\gamma_2}\right)^2=\left(\frac{139.2\ \text{h}}{15.0\ \text{h}}\right)^2=86.1184.$$

Furthermore, over any periodic treatment cycle $[0,T]$ returning $S_1(T)=S_1(0)$ and $S_3(T)=S_3(0)$, resistant subclone S_2 decays ($S_2(T)<S_2(0)$) whenever normalized kill impulses $\mu_i=k_{d,i}\langle C_i\rangle_T/r_1$ satisfy the Double-Bind Extinction Wedge $\mu_1<(1-1/\alpha_{21})+\mu_2/\alpha_{21}$, driving $S_2(0)=900\to0.0000$ over 5 cycles while maintaining $\min_t S_1(t)=S_{\text{floor}}=7812.5$.

3. Noisy Split-Mark Channel Capacity & Diachronic Clause-Bitrate Conservation

Theorem 3.1 (Antipodal KL Blind-Spot & Noisy Split-Mark Jeffreys Capacity). Under a stochastic DINA channel with slip $s=P(M_{j,c}=0\mid\eta_{j,c}=1)$ and guess $g=P(M_{j,c}=1\mid\eta_{j,c}=0)$, antipodal latent states $\eta=(1,0)$ and $\eta'=(0,1)$ induce identical scalar-sum distributions $P(S_j\mid1,0)\equiv P(S_j\mid0,1)$, yielding $D_{\text{KL}}(P(S_j\mid1,0)\parallel P(S_j\mid0,1))\equiv0$ for all $(s,g)\in[0,1]^2$. Conversely, the ordered Split-Mark vector $V_j=(M_{j,L},M_{j,R})$ achieves exact symmetric Jeffreys divergence:

$$J(V_j)=D_{\text{KL}}\big(P(V_j\mid1,0)\parallel P(V_j\mid0,1)\big)+D_{\text{KL}}\big(P(V_j\mid0,1)\parallel P(V_j\mid1,0)\big)=2(1-s-g)\log_2\left(\frac{(1-s)(1-g)}{sg}\right.$$

providing $J(V_j)=15.29$ bits/item at $s=g=0.05$ and 3-item mutual information $I(\Theta;Y^{(3)})=2.4193$ bits (85.6% Bayes accuracy vs 70.9% for scalar sum). Across 15 paired 19th-century (1818-1902) and 2026 C1/C2 English prose passages, per-word syntactic coding efficiency $\eta_{\text{syn}}=\text{PPS}/\bar{L}$ increases by $+53.14\%$ ($0.1101\to0.1687$ prop/word, $t(14)=+6.844$, $p=8.0\times10^{-6}$, Cohen's $d_z=+1.767$) while sentence-level propositional load remains conserved ($2.70\to2.92$, $p=0.4778$).